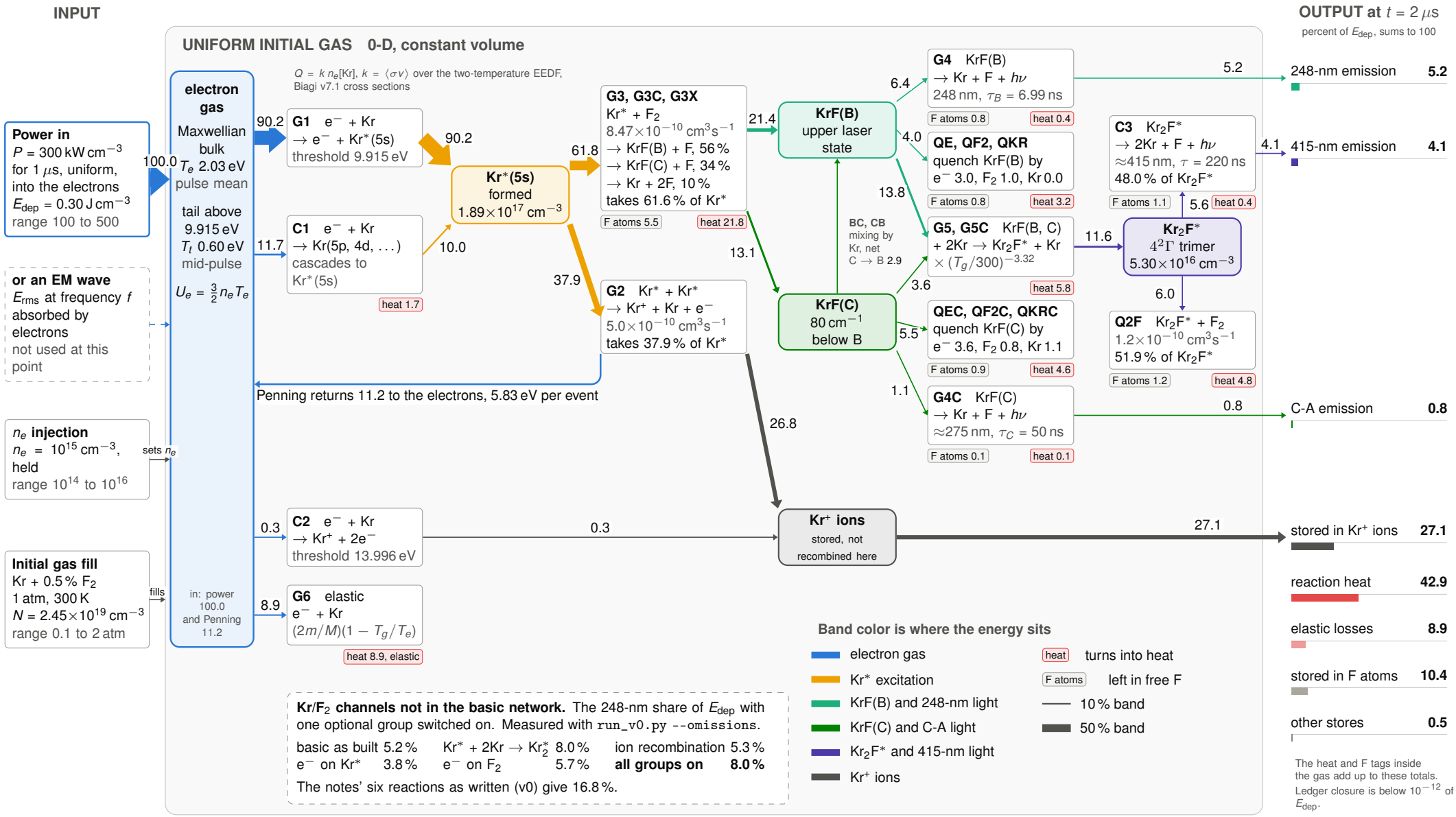
LIGHT system map, basic network: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e = 10^{15} cm^{-3} held, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 101.9 %, to Kr+ 27.1 %, to KrF(B) and KrF(C) 28.0 %, heat 42.9 %, elastic losses 8.9 %, emission 10.1 %. Of the KrF formed, 1.05×10^{17} cm^{-3}, 18.7 % radiates at 248 nm and 3.2 % in the C-A band, 50.5 % forms the trimer, and electrons, F_2 and Kr quench 19.0 %, 5.4 % and 3.2 %. B-C mixing carries 11.5 % of E_{dep} from B to C and 14.4 % back. At the end of the run T_g = 604 K and p = 2.03 atm, and F_2 fell below 10 % of the fill at 715 ns.

Generated from LIGHT v0 (0.1.0), basic network, by `docs/make_system_map_basic.py`. No number on this page is typed by hand. The rates and their sources are in `accuracy_overview.pdf`, Appendix A.